CSE 708
Graph Theory

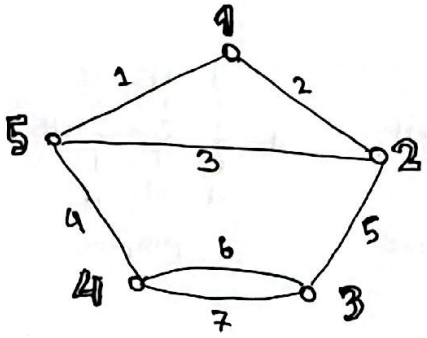
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$X = \text{mod}(21301002, 15) + 1$

$= 12$

2.12

Answer:



The above graph has 5 vertices, 1,2,3,4,5 and 7 edges 1,2,3,4,5,6,7

Adjacency Matrix A

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Incidence Matrix M

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$